

Bachelor of Technology — General Engineering - Data Analytics for Management and Finance

Table 1: General Engineering - Data Analytics for Management and Finance — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	41
	Discipline Electives	25
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-314*	Probability Theory and Statistics	3-1-0-4	4
2	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
3	HS-202	Principles of Economics	3-0-0-3	3
4	HS-307	Macroeconomics – I	3-0-0-3	3
5	IC-253	Programming and Data Structures	2-0-2-3	3
6	MA-222	Applied Mathematical Programming	3-1-0-4	4
7	MA-546	Introduction to Quantitative Finance		4
8	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
9	MA-654	Financial Engineering		3
10	MA-545	Technical Analysis of Financial Market		3
11	MA-579	Computational Methods in Financial Time Series Analysis		3
12	MB-311	Management Fundamentals for Engineers	3-0-0-3	3
13	MB-312	Economic Decision-Making for Engineers	3-0-0-3	3
Total				41

* Recoded w.e.f. batch 2025: DS-313 → DS-314 (Probability Theory and Statistics, 3-1-0-4 unchanged) and DS-413 → DS-416 (Fundamentals of Machine Learning, now 3-0-2-4).

Table 3: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-314*	Probability Theory and Statistics	3-1-0-4	4
4	DC	HS-202	Principles of Economics	3-0-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-307	Macroeconomics – I	3-0-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	MA-222	Applied Mathematical Programming	3-1-0-4	4
4	DC	MA-546	Introduction to Quantitative Finance		4
5	DC	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
2	DC	MA-654	Financial Engineering		3
3	DC	MB-311	Management Fundamentals for Engineers	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MB-312	Economic Decision-Making for Engineers	3-0-0-3	3
2	DC	MA-545	Technical Analysis of Financial Market		3
3	DC	MA-579	Computational Methods in Financial Time Series Analysis		3
4	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 25 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 11: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	DS-412	Matrix Computations for Data Science	4
2	HS-510	Essentials of Entrepreneurship	3
3	MA-211	Ordinary Differential Equations	4
4	MA-311	Mathematical Modelling	3

Table 12: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	DS-312	Computing Systems for Data Processing	4
2	DS-415	Optimization for Machine Learning	4
3	HS-481	International Economics	3
4	HS-504	Personal Finance and Portfolio Management	3
5	HS-550	Statistical Methods	4
6	MA-220	Partial Differential Equations	4
7	MA-221	Numerical Analysis	4
8	MA-524	Probability and Statistics	4
9	MA-552	Number Theory	3
10	MA-604	Introduction to Game Theory	3
11	ME-100	Reverse Engineering	1

Table 13: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Any Semester (Business)

Sl. No	Course Code	Course Name	Credits
1	MB-410	Marketing for Engineers	3
2	MB-411	Human Resource Management and Organizational Behaviour for Engineers	3
3	MB-412	Consumer Behaviour in Technology Markets	3
4	MB-413	Technology Entrepreneurship and Venture Creation	3
5	MB-414	Accounting Practices for Technology Businesses	3
6	MB-415	Financial Management for Technology Businesses	3
7	MB-416	Industrial Operations and Supply Chain Systems	3
<i>Compulsory project components — counted as DE, not selectable</i>			
8	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
9	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4